

Q1

What is the value of x^x ? (Assuming variable x is `unsigned int`)

☐ It depends on x

☒ Always 0

☐ Always 1

☐ Always `0xffffffff`

Q2

What is the value of $x^{\sim x}$ where variable x is of type `unsigned int`.

Note the expression contains `~` (bitwise negation) and not the `-` (minus/negative) operator.

☐ It depends on x

☐ Always 0

☐ Always 1

☒ Always `0xffffffff`
 $a \rightarrow 0/1$
 $\sim a \rightarrow 1/0$
 $a^{\sim a} = 1$

Q3

Consider the following code snippet,

```
char x = -2;
char y = x >> 1;
unsigned char z = ((unsigned char)x) >> 1;
```

Diagram illustrating the bit shifts:

- `x = -2` (binary: `1111 1111 1111 1111 1111 1111 1111 1100`)
- `y = x >> 1` (binary: `1111 1111 1111 1111 1111 1111 1111 1100`)
- `z = ((unsigned char)x) >> 1` (binary: `0111 1111`)

Q3.1

0.5 Points

After executing the code snippet, what is the value of `y` (please write a decimal number as your answer)

-1

Q3.2

0.5 Points

After executing the code snippet, what is the value of `z` (please write a decimal number as your answer)

127

Q4

Which value is the closest to $1 \ll 20$

- ☐ 1000
- ☒ 1 million
- ☐ 1 billion
- ☐ 2000
- ☐ 2 million
- ☐ 2 billion

$$2^{20} = (2^{10})^2 \approx 10^6$$

$$2^{10} = 1024 \approx 10^3$$

Q5

Q5 Bit-wise ops

1 Point

Variable x is of type unsigned int. Which of the following statements returns the most significant byte of x?

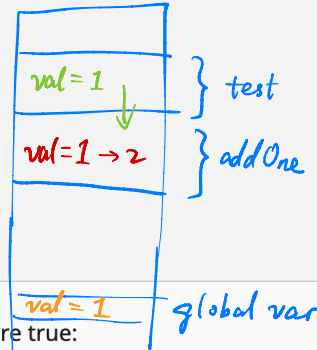
- ☐ (char)x
- ☒ (char)(x >> 24)
- ☐ (char)(x | 0xff000000)
- ☐ (char)(x & 0xff000000)
- ☐ None of the above



Q6

Consider the following code snippet,

```
1: void addOne(int val)
2: {
3:   val++;
4: }
5:
6: void test()
7: {
8:   int val = 1;
9:   addOne(val);
10:  assert(val == 2);
11:}
```

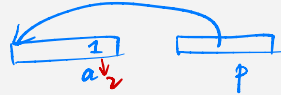


Which of the following statements are true:

- ☒ Running `test()` will result in assertion failure.
- ☐ Running `test()` will pass the assertion correctly.
- ☐ The `addOne` function argument `val` and the local variable `val` refer to the same variable
- ☒ The `addOne` function argument `val` and the local variable `val` are unrelated.
- ☐ The program will pass test correctly after moving line 8 out of the `test` function, making `val` a global variable.
- ☒ The program will fail the assertion after moving line 8 out of the `test` function, making `val` a global variable.

Q7

```
int a = 1;
int *p = NULL;
p = &a;
(*p)++;
printf("%d %d", *p, a);
```



Which are the possible printouts after running the above code snippet?

☐ 0 0

☐ 1 1

☒ 2 2

☐ 0 1

Q8

Q8.1 Pointers and arrays

0.5 Points

*int a[5] a: int **

Given variable definition `char *c[10];` what is the type of the expression `c[0]+1`?

*c: char ***

☐ `char **` *char **

☒ `char *`

☐ `char`

☐ none of the above

Q8.2

0.5 Points

Given variable definition `char *c[10];` what is the type of the expression `c+1`?

☒ `char **`

☐ `char *`

☐ `char`

☐ none of the above

Q8.3

0.5 Points

Given variable definition `char c[10];` what is the type of the expression `c[0]+1`?

*char **

☐ `char **` *char*

☐ `char *`

☒ `char`

☐ none of the above

Q8.4

0.5 Points

Given variable definition `char c[10];` what is the type of the expression `c+1`?

☐ `char **`

☒ `char *`

☐ `char`

☐ none of the above

Q9 Pointer Arithmetic

1 Point

Here's a C code fragment:

```
1: int x[3] = {1, 2, 3};  
2: int *p = x+1;  
3: _____  
4: printf("%d %d %d\n", x[0], x[1], x[2]);
```

$*(p+1) = 10$

$p[1] = 10$

$1[p] = 10$

In order for the above code fragment to output `1 2 10`, which of 1 line of code that you should put at Line-3?

☐ `p[0] = 10;`

☒ `p[1] = 10;`

☐ `p[2] = 10;`

☐ `*(p) = 10;`

☒ `*(p+1) = 10;`

☐ `*(p+2) = 10;`

☐ `p++;`

☐ `p--;`

$a[i] : *(a+i)$

$i[a]$

$1[p] = 10;$